portfolio.

S H O F A
J A N N A H

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About Me

A Master's graduate in Information Systems from Institut Teknologi Sepuluh Nopember, with two years of experience in IT with a strong interest in Data Analysis and Business Analysis with proven skills in Python and SQL. Experienced in data management, visualization, and deep learning.

Technical Skills

- Programming & Data Analysis: Python (including TensorFlow, PyTorch), SQL, Jupyter Notebook, Google Colab,
- Machine Learning and Deep Learning, Data Processing (Collection, Cleaning, Analysis)
- Data Visualization & NLP: Matplotlib, Seaborn, Text Preprocessing, Sentiment Analysis, Text Classification, Feature Engineering & Embeddings
- Tools & Productivity: Microsoft Lists, Power Apps, Trello, Microsoft Office (Word, Excel, PowerPoint)

Education & Certifications

🎓 Bachelor's Degree in Information Systems – Universitas Dinamika, 2021
🎓 Master's Degree in Information Systems – Institut Teknologi Sepuluh Nopember, 2025
📜 Certifications:

- Google Data Analytics by Coursera
- IBM Data Analyst by Coursera
- Machine Learning by DeepLearning AI by Coursera
- Data Analytics by LinkedIn Learning
- Data Visualization with Tableau by Coursera

Publications & Conferences

📄 Visual Aspect-Based Sentiment Analysis for Tourist Reviews Using Deep Learning Method, IEEE 2024 International Conference on Information Technology Systems and Innovation (ICITSI), Bandung, Indonesia.
📄 Exploring Deep Learning Architectures for Effective Visual Question Answering, IEEE 2024 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT), Sakhir, Bahrain.
📄 Exploring Semantic Similarity among MUI Fatwas: A Computational Analysis using Generalized Jaccard Similarity, Halal Research Journal, Vol. 4, No. 2, pp. 78-88, 2024.

Projects & Research

Deep Learning & NLP

🎓 **Master's Thesis: Aspect-Based Multimodal Sentiment Analysis for Tourist Reviews using Deep Learning**

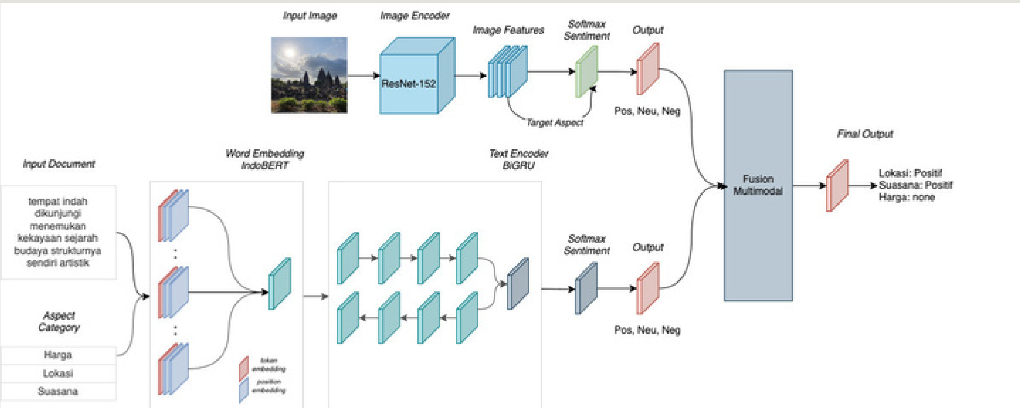
- Developed a deep learning model for multimodal data analysis, integrating computer vision for image processing and natural language processing (NLP) for text analysis across three aspects: location, price, and atmosphere.
- Used ResNet152, Inception, and DenseNet for image feature extraction and LSTM, BiGRU, and CNN-BiLSTM for text processing with IndoBERT.
- Applied late fusion via concatenation to enhance sentiment classification across multiple aspect.

Results

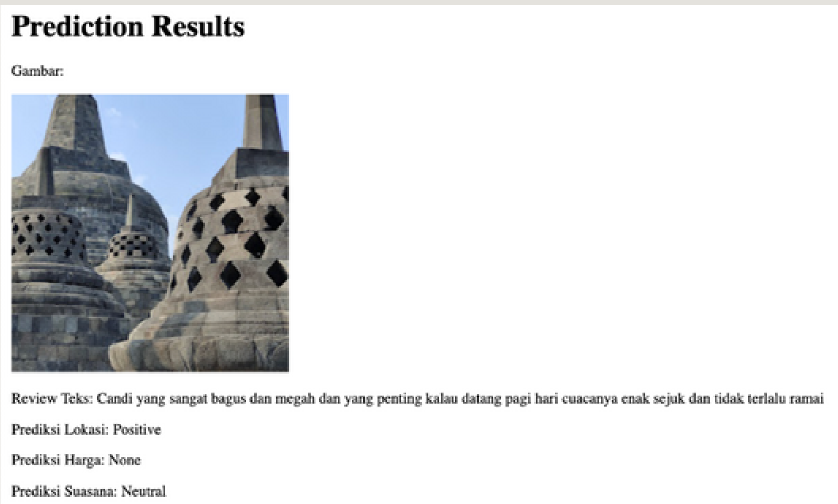
🏆 Best Model: CNN-BiLSTM + ResNet-152 (0.98 accuracy!) – Residual connections helped align text & image features effectively.

📉 InceptionV3: Strong for images, but struggled in multimodal sentiment tasks.

🤖 RNN + DenseNet-121: Too complex for fusion, leading to lower accuracy than text-only models.



Architecture for ABMSA



Results of ABMSA

- **<u>Semantic Similarity using Generalized Jaccard Similarity</u>**

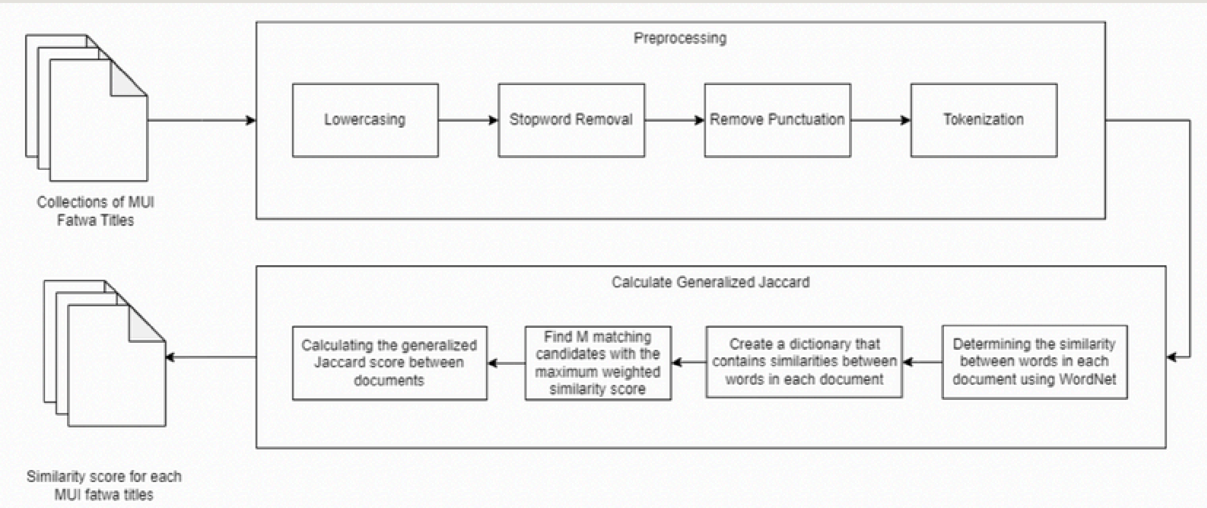
- Developed a semantic similarity model to analyze MUI fatwa documents, addressing challenges in overlapping fatwas.
- Demonstrated a 25.86% improvement over the standard Jaccard method in string matching quality.
- Used Python and NLP libraries for preprocessing, tokenization, and similarity scoring.

Results

✅ 73 fatwa documents had a similarity score ≥ 0.5, showing significant resemblance.
❌ 77,028 documents had scores < 0.5, indicating lower similarity or distinct topics.
⚖️ Findings emphasize the need for better classification in Islamic legal discourse.

$$GJ\ (x, y) = \frac{\sum(x_i, y_j \in M^{s(x_i, y_j)})}{|B_x| + |B_y| - |M|}$$

Formula of Generalized Jaccard Similarity



Overal research flow

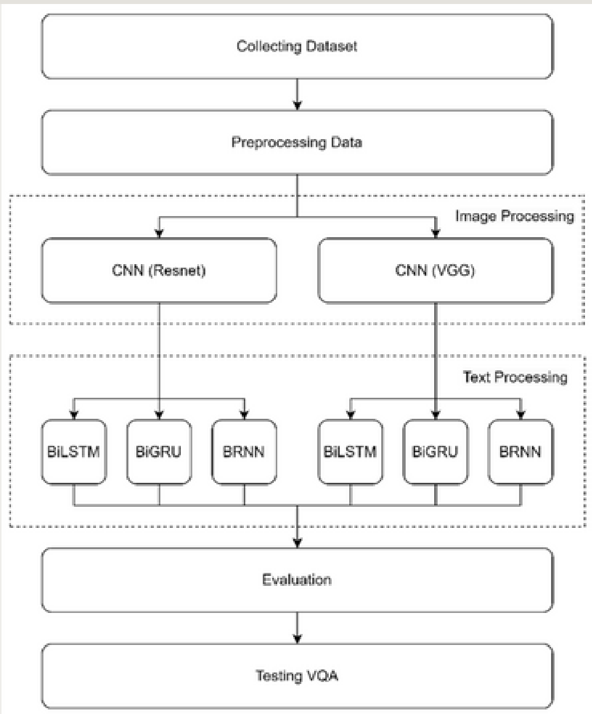
Fatwa 1	Fatwa 2	Similarity GJS	Similarity JS
['fatwa', 'penyerangan', 'amerika', 'serikat', 'sekutunya', 'irak']	['penyerangan', 'amerika', 'serikat', 'sekutunya', 'irak']	0.83	0.83
['pandangan', 'ruu', 'larangan', 'minuman', 'beralkohol']	['ruu', 'larangan', 'minuman', 'beralkohol']	0.8	0.8
['pelaksaan', 'shalat', 'jumat', 'dzikir', 'kegiatan', 'keagamaan', 'tempat', 'masjid']	['pelaksanaan', 'shalat', 'jumat', 'dzikir', 'kegiatan', 'keagamaan', 'tempat', 'masjid']	0.78	0.78
['hukum', 'alkohol', 'minuman']	['hukum', 'alkohol']	0.67	0.67
['pencurian', 'energi', 'listrik']	['aliran', 'ahmadiyah']	0.25	0
['panduan', 'penyelenggaraan', 'ibadah', 'bulan', 'ramadan', 'syawal', '1442']	['arah', 'kiblat']	0.125	0

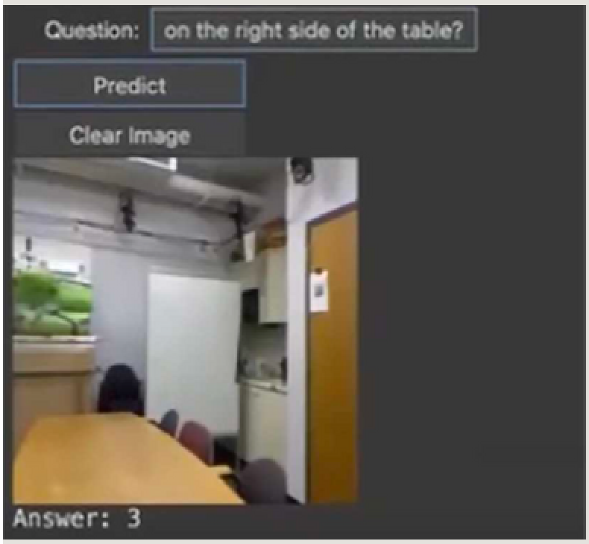
The similarity results between fatwas with Generalized Jaccard Similarity score

🧠 Exploring Deep Learning Architectures for Effective Visual Question Answering

- Compared different deep learning architectures for Visual Question Answering (VQA) to determine the most effective model.
- Image Feature Extraction: Compared ResNet152 and VGG16 for high-level visual representation.
- Text Processing: Implemented BiLSTM, BiGRU, and BRNN for better question understanding.
- Fusion Strategy: Applied concatenation to combine image feature vectors with BRNN-based text representations. Mapped high-level fused features through non-linear transformations. Used a softmax classification layer to predict answers based on the dataset.

Results:
🥇 VGG16 + BiGRU was the best performer with 55.89% validation accuracy and 52.30% test accuracy.
🥈 ResNet + BiLSTM had the lowest accuracy 25.12% validation, 23.58% test.
🔍 These findings help enhance VQA models for better AI-driven visual understanding!

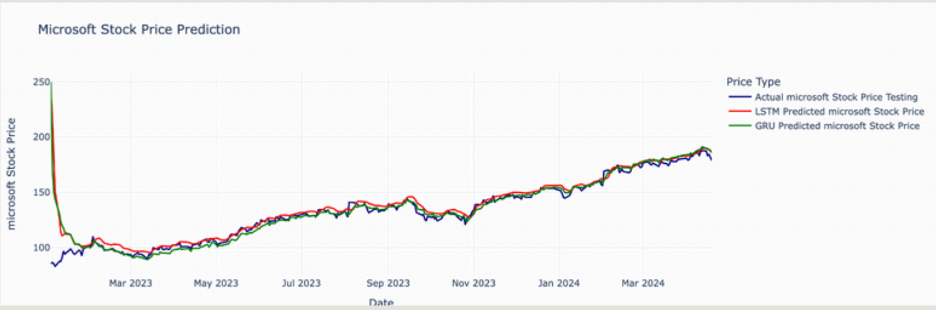




Machine Learning & Forecasting

📈 <u>Stock Forecasting using LSTM and GRU</u>

💰Developed deep learning models to forecast stock prices for five blue-chip stocks: Apple (AAPL), Microsoft (MSFT), Google (GOOGL), Amazon (AMZN), and Tesla (TSLA).
🧠 Utilized LSTM and GRU architectures to analyze historical stock data and improve prediction accuracy.
📊 Evaluated model performance using RMSE, MAPE, and MAE to assess forecasting reliability.
⚡ LSTM and GRU models provide reasonably accurate close stock price predictions, despite external factors like economic and political events.



Prediction results on the test data for Google and Microsoft stocks

10-day future prediction for Microsoft stock

Models		MSE	MAE	MAPE
Google	LSTM	13.72	3.002	0.023 (2.3%)
	GRU	13.95	3.045	0.024 (2.4%)
Amazon	LSTM	61.23	4.75	0.035 (3.5%)
	GRU	41.50	4.72	0.020 (2.0%)
Apple	LSTM	22.93	3.89	0.022 (2.2%)
	GRU	115.47	10.24	0.061 (6.1%)
Microsoft	LSTM	201.04	5.01	0.034 (3.4%)
	GRU	157.97	4.37	0.031 (3.1%)
NVIDIA	LSTM	11603.30	61.38	0.120 (12.0%)
	GRU	3830.488	45.17	0.094 (9.4%)

MSE, MAE, and MAPE evaluation metrics for the five stocks

Data Visualization & Business Intelligence

🔷 Customer Analysis using Tableau

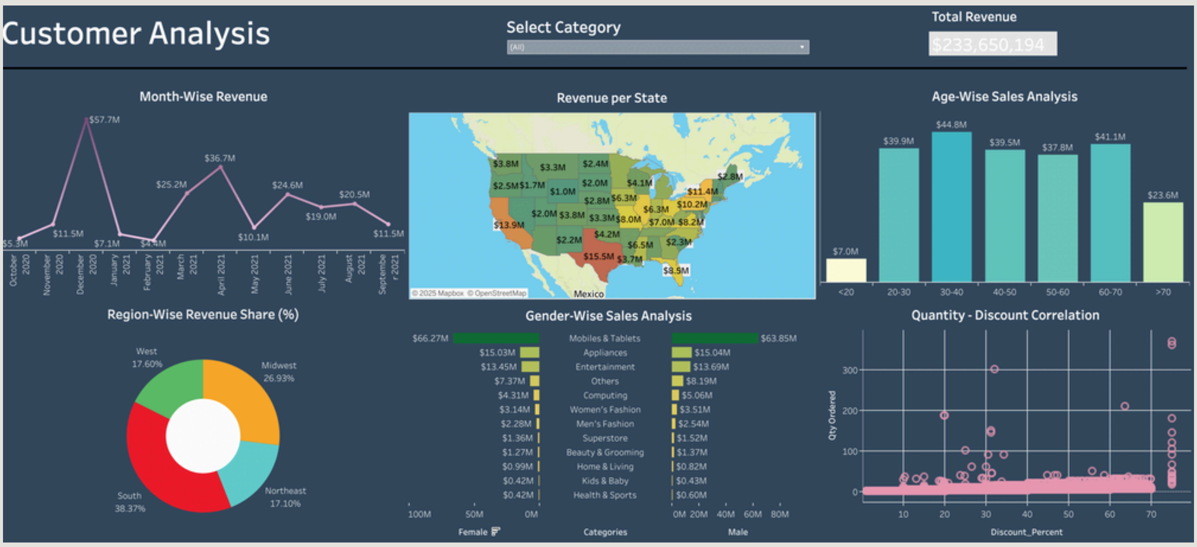
This interactive dashboard provides a comprehensive analysis of customer revenue and sales trends. Built with data visualization techniques, it helps businesses gain insights into revenue distribution across different dimensions.

🔥 Key Features & Insights

- Month-Wise Revenue Trend – Line chart showing monthly revenue fluctuations.
- Revenue per State – A geographic heatmap visualizing state-wise revenue distribution.
- Age-Wise Sales Analysis – A bar chart displaying sales performance across different age groups.
- Region-Wise Revenue Share – A donut chart highlighting revenue percentage by region.
- Gender-Wise Sales Analysis – A stacked bar chart comparing sales categories by gender.
- Quantity-Discount Correlation – A scatter plot showing the relationship between quantity ordered and discount percentage.

🚀 Impact

- Identifies peak sales periods for better inventory planning.
- Highlights high-performing states and regions for strategic expansion.
- Shows the impact of discounts on sales volume to optimize pricing strategies.



🔹 Amazon Prime Movies Dashboard using Tableau

This dashboard was created using Tableau to analyze and visualize data related to movies and TV shows available on Amazon Prime Video. The goal was to provide a comprehensive overview of the platform's content, highlighting key trends and insights.

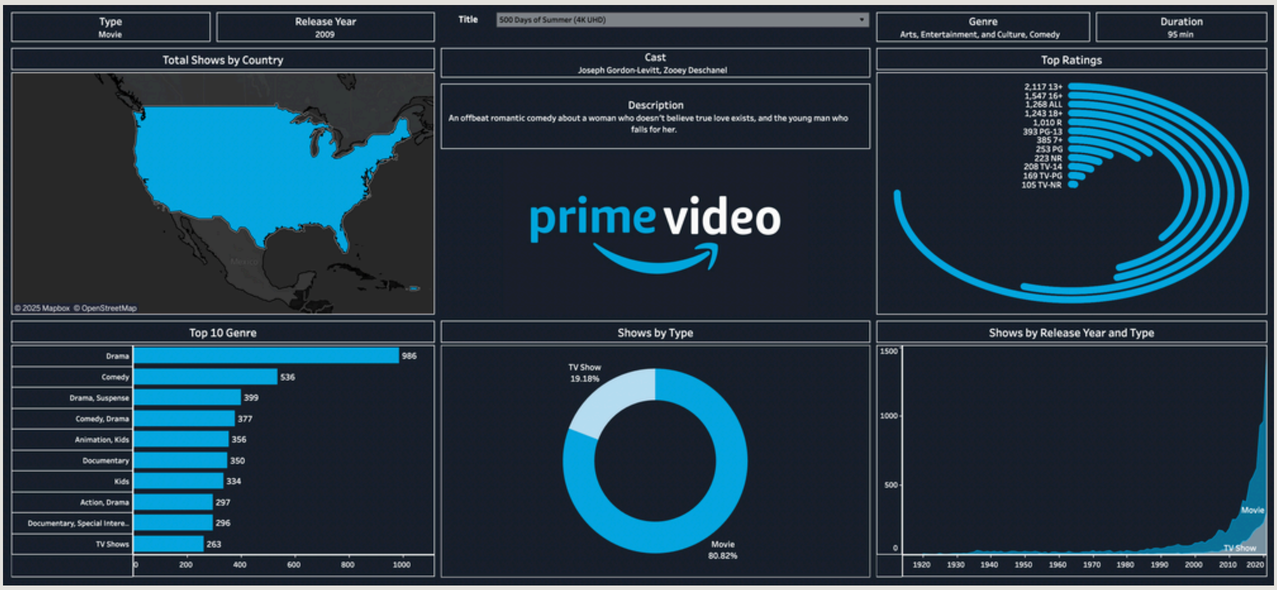
🔥 Key Features & Insights

- Shows by Country – See where Amazon Prime's content is most available.
- Top 10 Genres – Discover the most popular movie and TV show categories.
- Movies vs. TV Shows – A quick breakdown of content types.
- Release Trends – Track how content production has changed over the years.
- Top-Rated Content – Find the highest-rated movies and shows instantly.
- Movie Details – Click to view key info about a specific movie.
- Fully interactive! Filter, explore, and uncover hidden trends in Amazon Prime Video's library.

🚀 Impact:

✅ Content Strategy Optimization – Helps Amazon Prime and content creators identify popular genres & regional preferences to improve content curation.

✅ User Engagement Insights – Provides data-driven insights into what audiences watch most, influencing recommendations & marketing strategies.



🔹 Super Store Sales Dashboard using Tableau

This dashboard was created using Tableau to analyze sales data for a "Super Store," providing insights into key performance indicators and trends. The goal was to create a comprehensive overview that allows for data-driven decision-making.

✨ Key Features & Insights:

- Performance Metrics – Snapshot of Total Sales, Quantity Sold, & Profit
- Sales & Profit by State – Heatmap highlighting top & low-performing regions
- Sales by Segment – Donut chart for Consumer, Corporate, & Home Office sales distribution
- YoY Monthly Sales – Line chart comparing 2019 vs. 2020 trends
- Ship Mode Sales – Breakdown of Standard, Express, & Same Day shipping
- Payment Mode – Donut chart of Cards, Online, & COD payments
- YoY Profit Trends – Monthly profit analysis across two years
- Category & Sub-Category Sales – Best-selling products in Office Supplies, Tech, & Furniture

🚀 Impact:

✅ Helps identify top-selling products & customer preferences
✅ Optimizes shipping strategies & sales performance
✅ Enables data-driven decision-making for better profitability

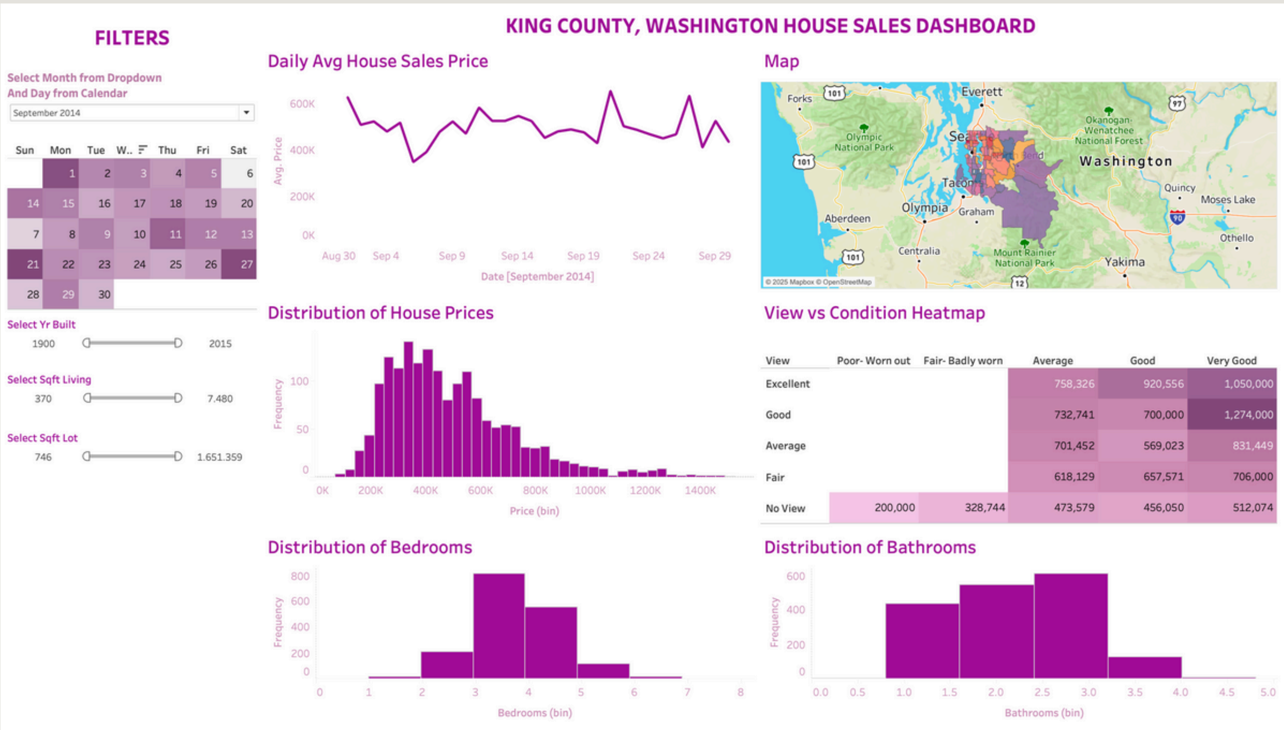
🔹 **King County, Washington House Sales Dashboard**

This interactive dashboard provides insights into house sales in King County, Washington, using visualizations and filters to analyze key trends in housing prices, property attributes, and geographic distribution.

✨ Key Features & Insights:

- Displays the fluctuation of average house prices over time from 2014 to 2015.
- Maps house sales across King County with color-coded density regions.
- Shows the frequency of house sales across different price ranges.
- Compares house condition ratings (Poor, Fair, Average, Good, Very Good) with views (Excellent, Good, Average, Fair, No View) to analyze price trends.
- Visualizes the frequency of house sales by the number of bedrooms.
- Displays the distribution of properties based on the number of bathrooms.

🚀 Impact:
✅ Helps buyers, sellers, and real estate agents understand price trends and property value.
✅ Provides clarity on how property attributes (view, condition, size) influence pricing.
✅ Enables quick comparisons and targeted property analysis.



🔹 **Hospitality Data Analysis & Visualization Project**

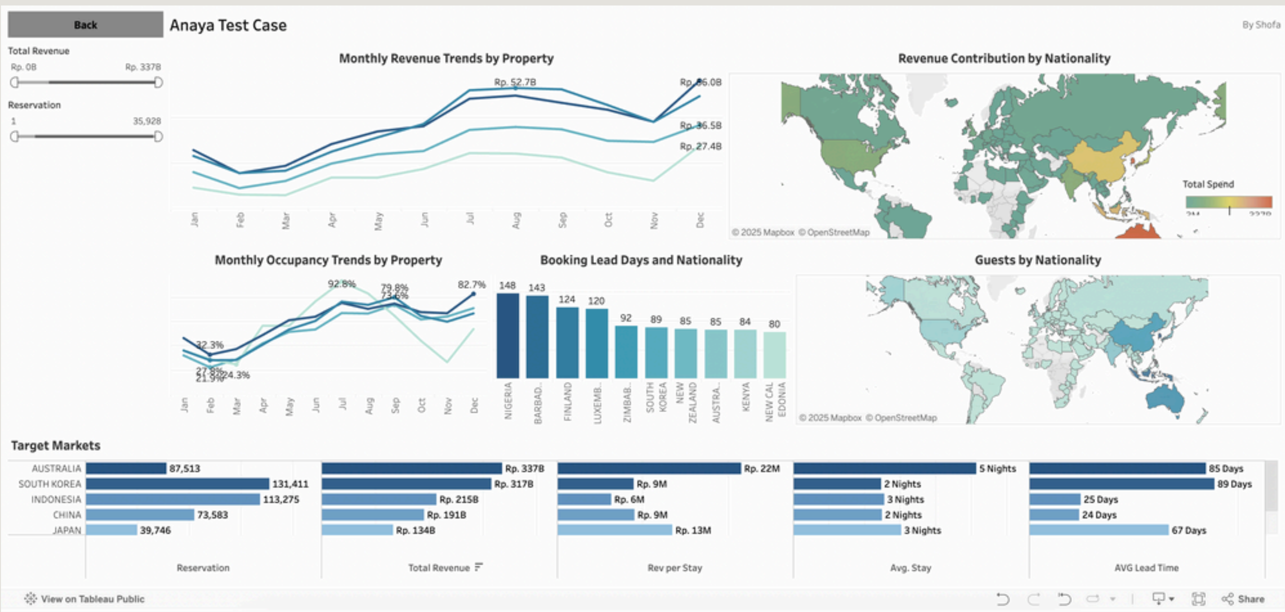
This interactive dashboard provides insights into key hospitality metrics such as revenue, occupancy, booking behavior, and guest demographics—designed to support data-driven decision-making in the hotel industry.

✨ Key Features & Insights:

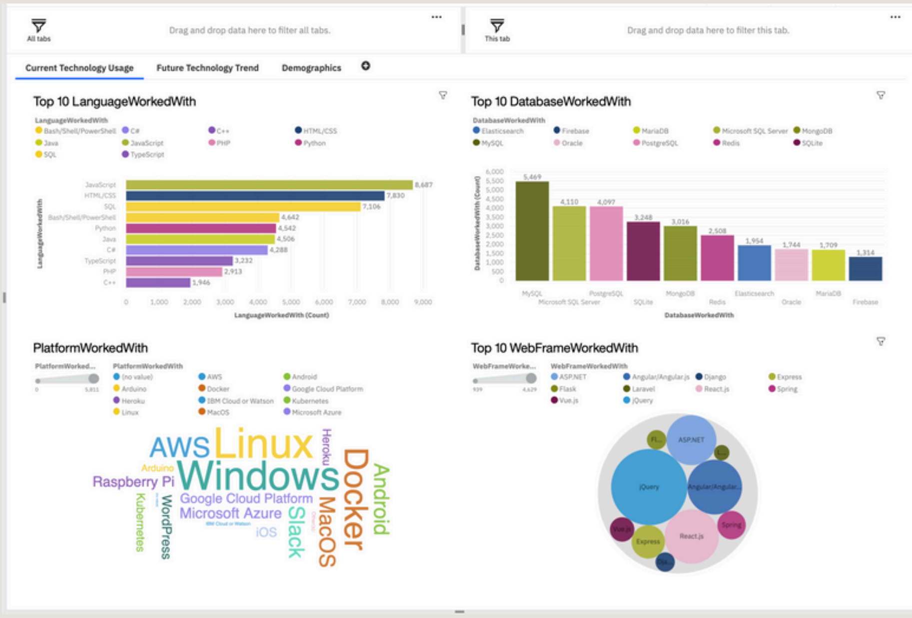
- Monthly Performance Dashboard
  - Visualizes revenue and occupancy trends by property to monitor operational health and seasonal patterns.
  - Booking Behavior Analysis
- Shows relationships between:
  - 🧳 Lead time and nationality (booking habits)
  - 🛌 Length of stay and total guest spend (customer value)
- 🌍 Geographical Insights
  - Maps and charts highlighting guest origins
  - Revenue contribution by nationality
- Identification of high-value target markets based on historical trends

🚀 Impact:

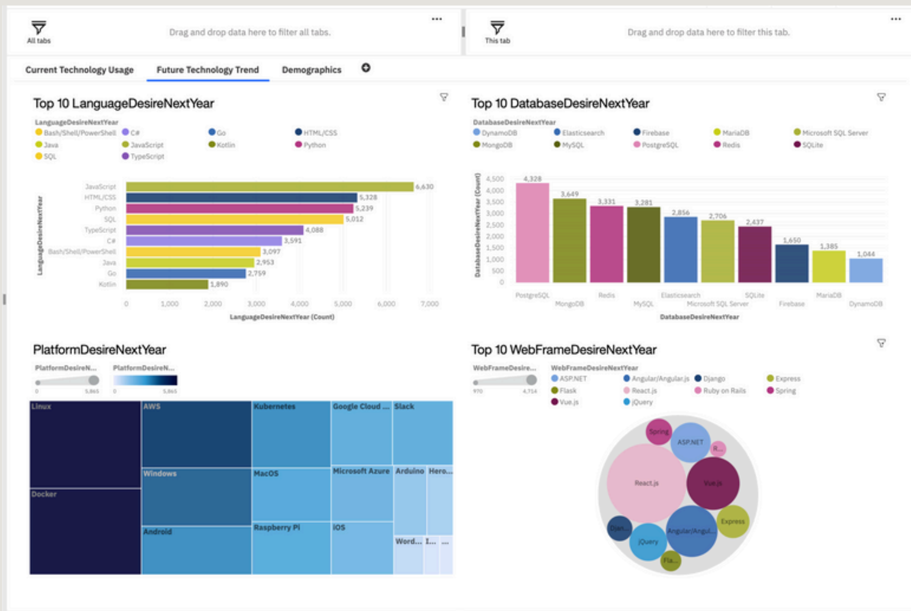
✅ Simplified complex data into actionable insights for business users
✅ Enabled better forecasting and promotional planning
✅ Highlighted guest segments with the highest ROI potential
✅ Improved stakeholder understanding of revenue drivers and occupancy trends



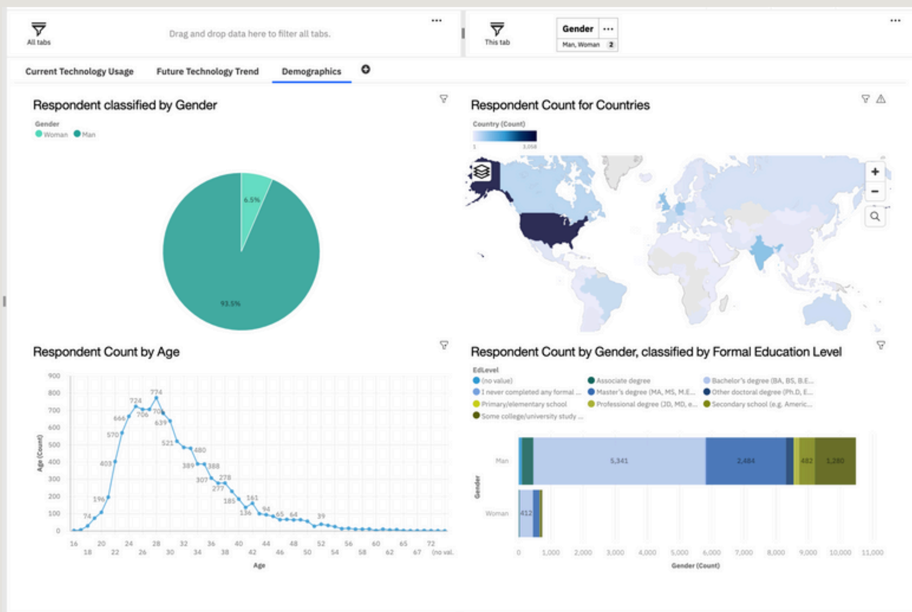
Dashboard IBM Data Analytics Capstone Project using IBM Cognos



Interactive insights into tech trends!



Future Tech Trends: JavaScript, PostgreSQL, Linux, and React.js lead next-year adoption!



Demographics: 93.5% men, majority aged 25-34, highest respondents from the US, most hold a Bachelor's degree.

thank you

shofa jannah